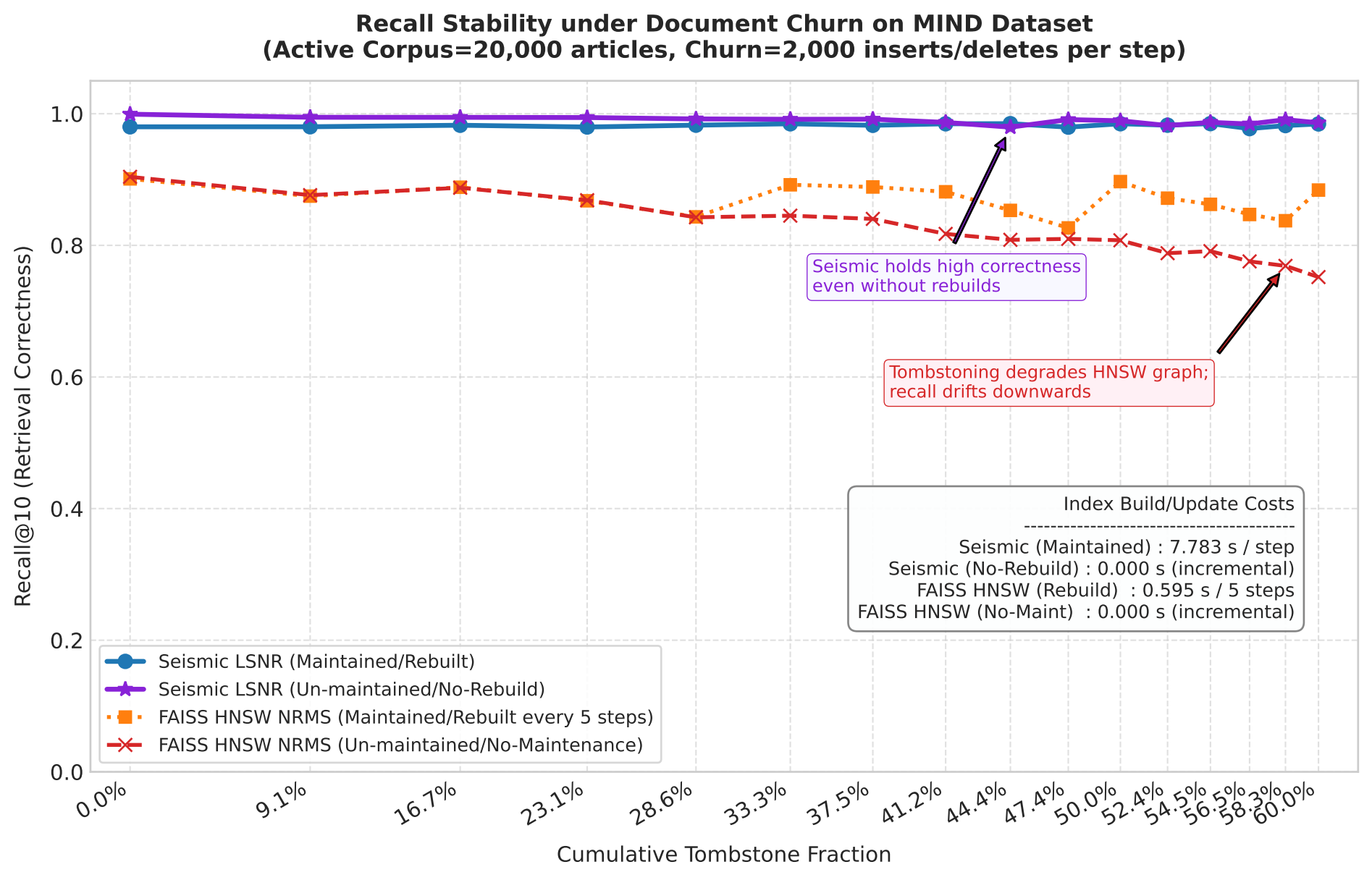
Recall Stability under Document Churn on MIND Dataset
(Active Corpus=20,000 articles, Churn=2,000 inserts/deletes per step)
Recall@10 (Retrieval Correctness)
1.0
0.8
0.6
0.4
0.2
0.0
0.0%
9.1%
16.7%
23.1%
28.6%
33.3%
37.5%
41.2%
44.4%
47.4%
50.0%
52.4%
54.5%
56.5%
58.3%
60.0%
Cumulative Tombstone Fraction
Seismic holds high correctness
even without rebuilds
Tombstoning degrades HNSW graph;
recall drifts downwards
Index Build/Update Costs
Seismic (Maintained) : 7.783 s / step
Seismic (No-Rebuild) : 0.000 s (incremental)
FAISS HNSW (Rebuild) : 0.595 s / 5 steps
FAISS HNSW (No-Maint) : 0.000 s (incremental)
Seismic LSNR (Maintained/Rebuilt)
Seismic LSNR (Un-maintained/No-Rebuild)
FAISS HNSW NRMS (Maintained/Rebuilt every 5 steps)
FAISS HNSW NRMS (Un-maintained/No-Maintenance)